Research Plan

Robust Temporal Representation Learning for Weakly Supervised Video Anomaly Detection

A doctoral research plan focused on efficient temporal models, vision-language semantics, and generalization to unseen anomalies.

Presented to the doctoral jury: Máximo Enrique Pacheco Martínez

WSVAD | SSMs | VLMs

Defense Roadmap

01 Academic background

02 Context and research problem

03 Scientific gap, objective and central hypothesis

04 Three-stage methodology, previous research, evaluation, and expected contributions

Academic Background

Maximo Enrique Pacheco Martinez | Mexican | BUAP | Computer science researcher with work in artificial intelligence particularly in metaheuristics and NLP.

Profile

My previous work provided a strong foundation in machine learning, representation learning, and sequence modeling. These experiences motivated my transition from Natural Language Processing to Computer Vision, where temporal representation learning has become the main focus of my PhD.

Publications

Binary harmonic search for **attribute selection**. Research in Computing Science, Ciudad de México, México, August 2022

Harmony search for **hyperparameters optimization** of a low resource language **transformer** model trained with a novel parallel corpus Ocelotl **Nahuatl – Spanish.** Systems and Soft Computing, Netherlands, December 2024

Undergraduate thesis

Recurrent neural networks applied **to speech recognition** and **machine translation** from Spanish to Japanese (2019)

Master's thesis

Machine translator from Nahuatl to Spanish (2023)

Scholarships and awards

CONCYTEP Young Researchers Program, 2014

CONCYTEP Thesis Scholarship, 2018

CONACYT Graduate Scholarship, 2021-2023

MEXT – Research - Current

Conference Work

Binary Harmony Search for Attribute Selection, CONACIC (2022)

Ocelotl translator nahuatl – spanish, 4th International Conference on Natural Language Processing for Indigenous Languages (2024)

Ocelotl parallel corpus, 5th International Conference on Natural Language Processing for Indigenous Languages (2025)

What is Video Anomaly Detection (VAD)?

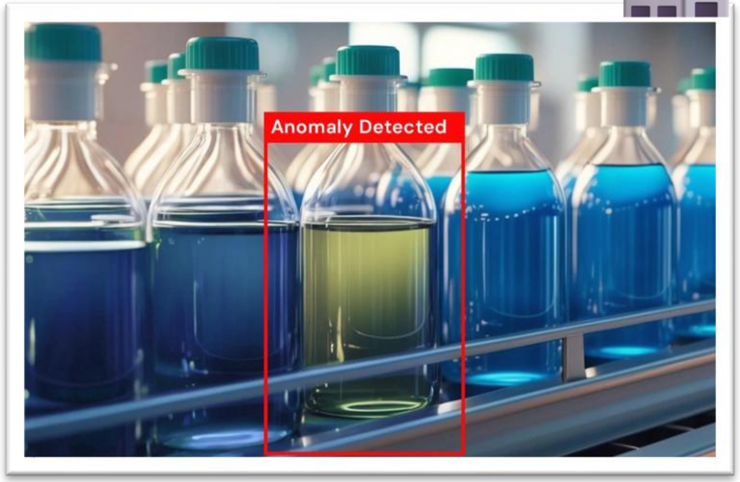
VAD

Computer vision task that aims to **identify events** in videos that significantly **differ** from learned **normal patterns**. Its main **challenge** is the **rarity and diversity of anomalous events**, making abnormal behavior difficult to model comprehensively.

Why Video Anomaly Detection (VAD) Matters?

Applications

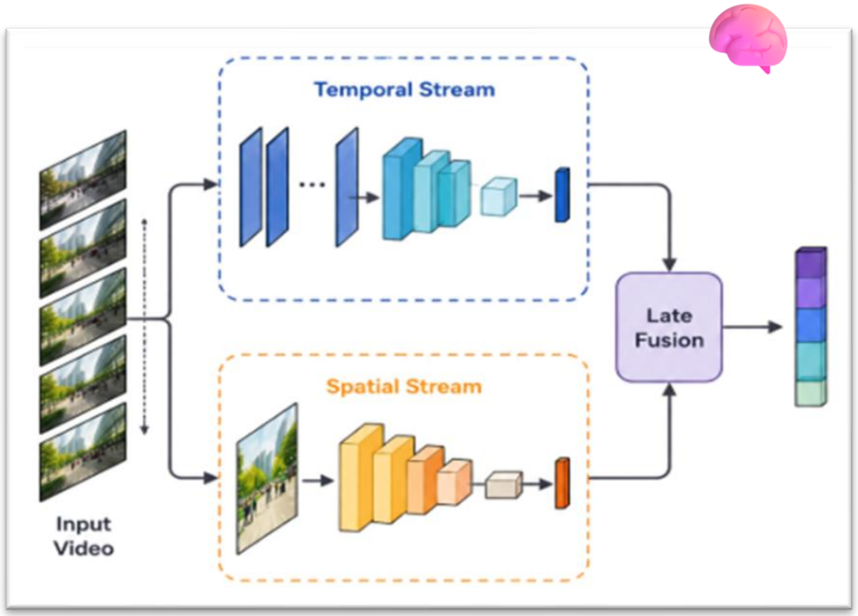
- Intelligent surveillance for **public safety** 📹
- Industrial inspection 🏭
- Autonomous systems 🚗

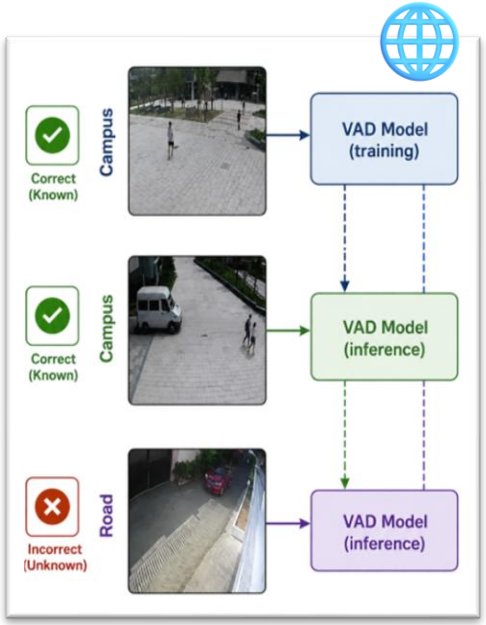


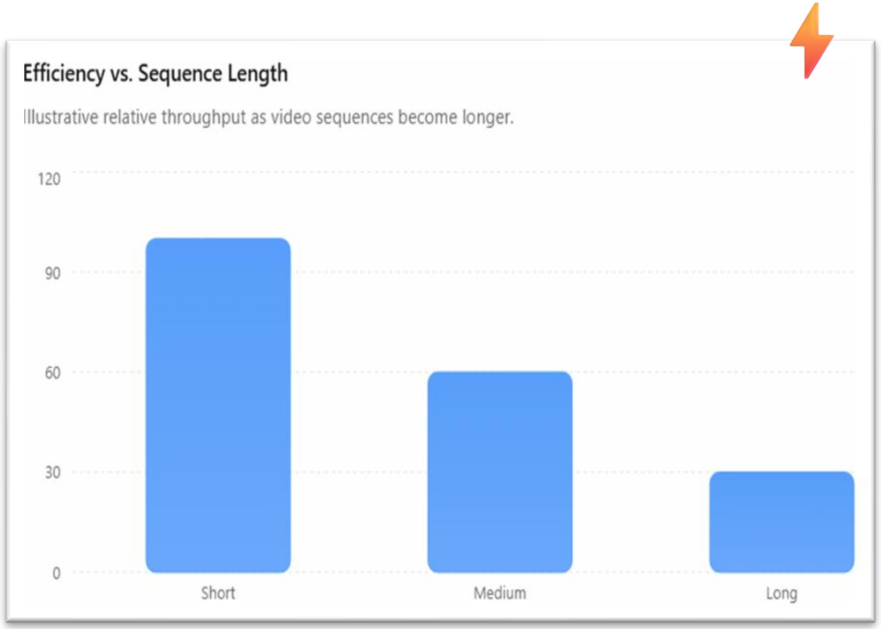
Why Video Anomaly Detection (VAD) Matters?

Challenges

- Learning **effective** temporal representations 🧠
- **Generalizing** to unseen anomalies 🌐
- **Efficiently** modeling long video sequences ⚡





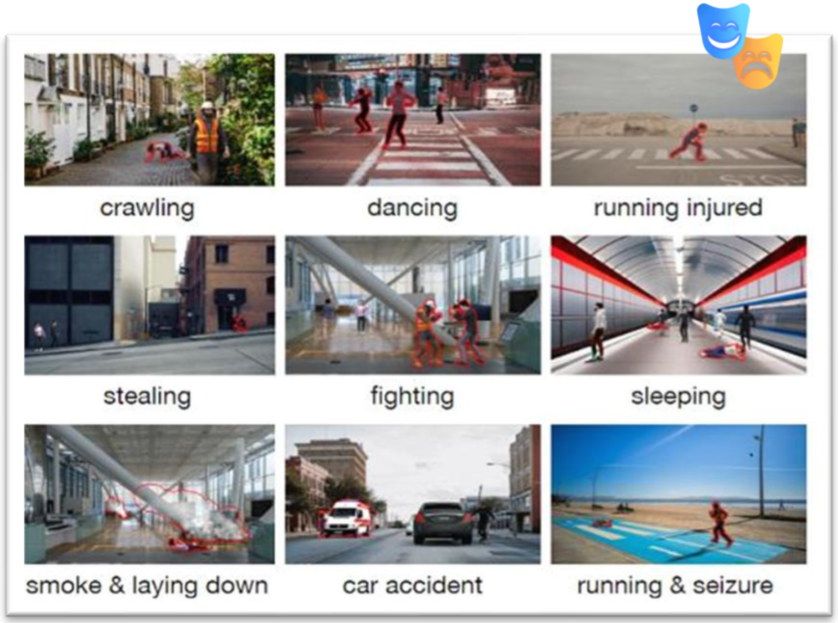


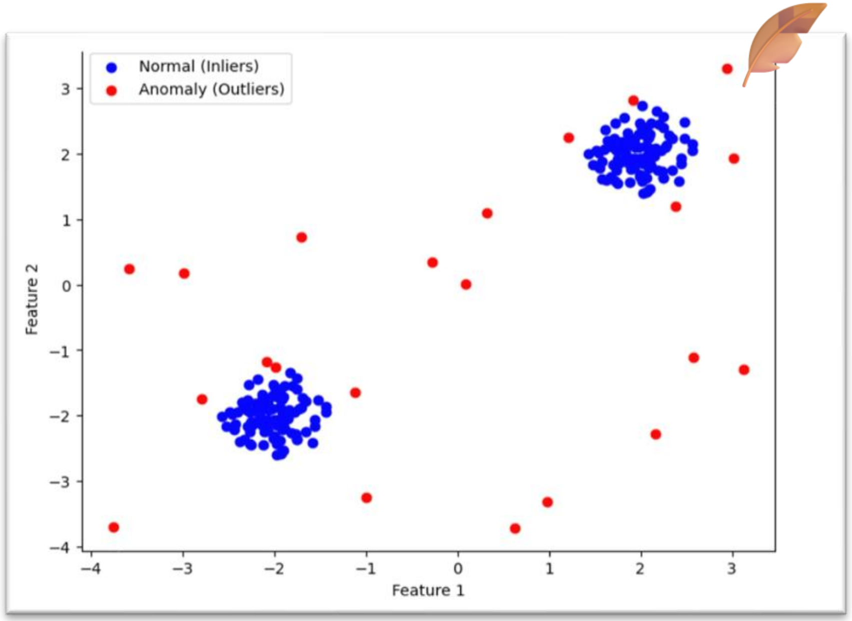
Why Video Anomaly Detection (VAD) Matters?

Practical Difficulty

Anomalies are:

- Diverse 🎭
- Ambiguous ❓
- Rare 🪶





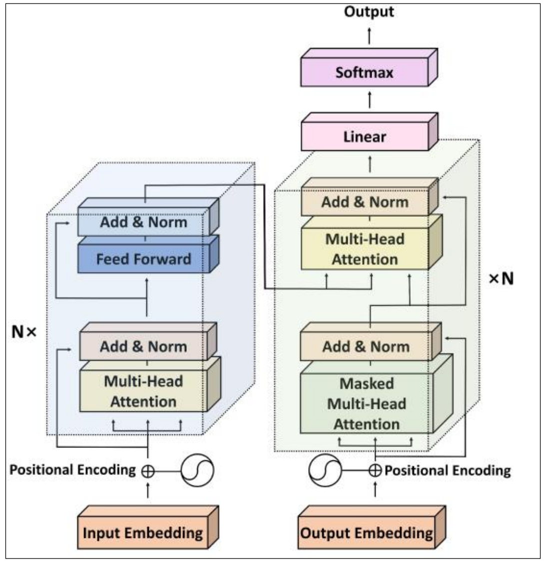
Why Video Anomaly Detection (VAD) Matters?

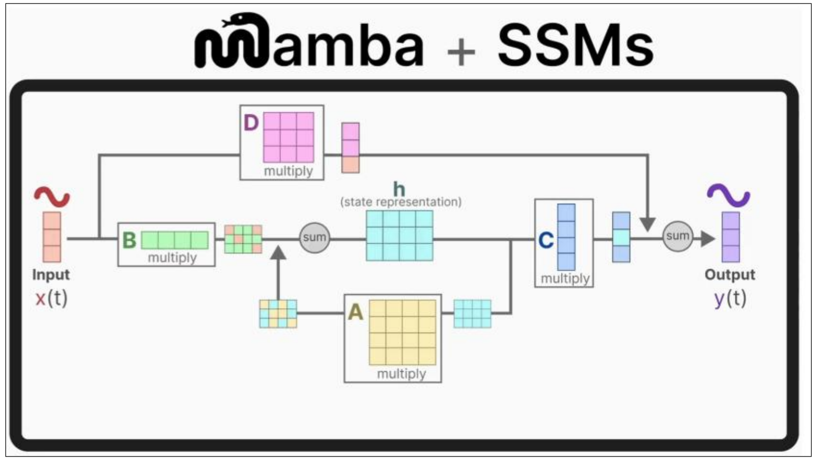
🟧 🎭 **Diverse · ❓ Ambiguous · 🪶 Rare → 🛡️ Temporal & Semantic Robustness**

Weak Supervision: In WSVAD, **only video-level labels** are available during training, making the task even more **challenging** because the model must **infer** which temporal **segments** are responsible **for** the **anomalous** event.

Research Gap

- Transformers and VLM-based methods have progressed, but may be costly for long videos.
- SSMs such as Mamba promise efficiency, yet their original causal design does not directly fit VAD.





The starting point is not a single architecture; rather, it is determining which temporal properties the representation should learn and which architecture can best improve the evaluation metrics.

Objective

Investigate how robust temporal representations improve anomaly detection in long and complex videos.

Central Hypothesis

Temporal representations enriched with complementary semantic or contextual information (e.g. VLM) can improve anomaly detection performance.

If temporal representations effectively integrate complementary semantic or contextual information, they can better distinguish normal from anomalous patterns, improving the detection of weak, ambiguous, or previously unseen anomalies.

Efficiency

Long sequences without quadratic cost.

Semantics

Anomaly concepts beyond closed classes.

Robustness

Better behavior under unseen events and domains.

Three-Stage Methodology

The plan moves from understanding the state of the art to semantic-temporal integration.

01

Comparative Study

Analyze CNN, Transformer, and VLM approaches for VAD/WSVAD; review RTFM, VadCLIP, OV-VAD, and related work.

- Identify strengths, limitations, and temporal modeling strategies.

02

Efficient Temporal Modeling

Investigate SSMs, particularly Mamba and its variants as alternatives to Transformers.

- Characterize their temporal representations and identify when they outperform attention-based models.

03

Semantic Integration

- Combine SSM-based temporal representations with complementary semantic or contextual information (e.g., from Vision-Language Models) to improve the recognition of weakly defined or previously unseen anomalies.

Specific Limitations in Previous Research

Previous approach	Strength	Remaining problem	Research direction
RTFM	Robust MIL + temporal dependencies	Temporal modeling relies on CNN/self-attention	**Efficient sequence modeling**
VadCLIP	Rich visual-language semantics	Limited temporal reasoning of dynamic anomalies	**Temporal-semantic integration**
OV-VAD	Detects unseen anomaly categories	Open-vocabulary semantics ≠ robust temporal representation	**Generalizable temporal representation**
Mamba-based VAD	Efficient long-sequence modeling	Original causal formulation is not directly suited to VAD	**VAD-specific non-causal SSM design**

Robust Temporal Representation Learning:

1. Efficient temporal modeling→ SSM/Mamba-based sequence modeling
2. Semantic enrichment→ VLM-based contextual/semantic information
3. Robust generalization→ representation designed to handle weak, ambiguous, and unseen anomalies

WSVAD

Experimental Evaluation

The method will be defended through accuracy, efficiency, and robustness on established benchmarks.

AUC

Frame-level temporal accuracy

AP

Quality under class imbalance

Cost

Memory, runtime, and scalability

Robustness

Generalization to unseen anomalies

Datasets UCF-Crime | XD-Violence | ShanghaiTech

Expected Contributions

The outcome aims to contribute both scientific insight and practical methods for WSVAD.

Scientific Contribution

- A better understanding of temporal representation learning for Weakly Supervised Video Anomaly Detection.
- Combining efficient sequence modeling with semantic information.

Practical Impact

- More robust, scalable, and generalizable video anomaly detection systems under weak supervision.

Thank you for your attention

References

[1] Sultani, W., Chen, C., & Shah, M. (2018). Real-world anomaly detection in surveillance videos. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 6479–6488.

[2] Tian, Y., Pang, G., Chen, Y., Singh, R., Verjans, J. W., & Carneiro, G. (2021). Weakly-supervised video anomaly detection with robust temporal feature magnitude learning. Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 4975–4986.

[3] Wu, P., Zhou, Y., Wang, X., et al. (2024). VadCLIP: Adapting vision-language models for weakly supervised video anomaly detection. Proceedings of the AAAI Conference on Artificial Intelligence, 38(6), 6076–6084.

[4] Wu, P., Zhou, Y., Wang, X., et al. (2024). Open-vocabulary video anomaly detection. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 18267–18276.

[5] Gu, A., & Dao, T. (2024). Mamba: Linear-time sequence modeling with selective state spaces. International Conference on Learning Representations (ICLR).

[6] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems (NeurIPS), 30.

[7] Acsintoae, A., Florescu, A., Georgescu, M. I., et al. (2022). UBnormal: New benchmark for supervised open-set video anomaly detection. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 20157–20167.